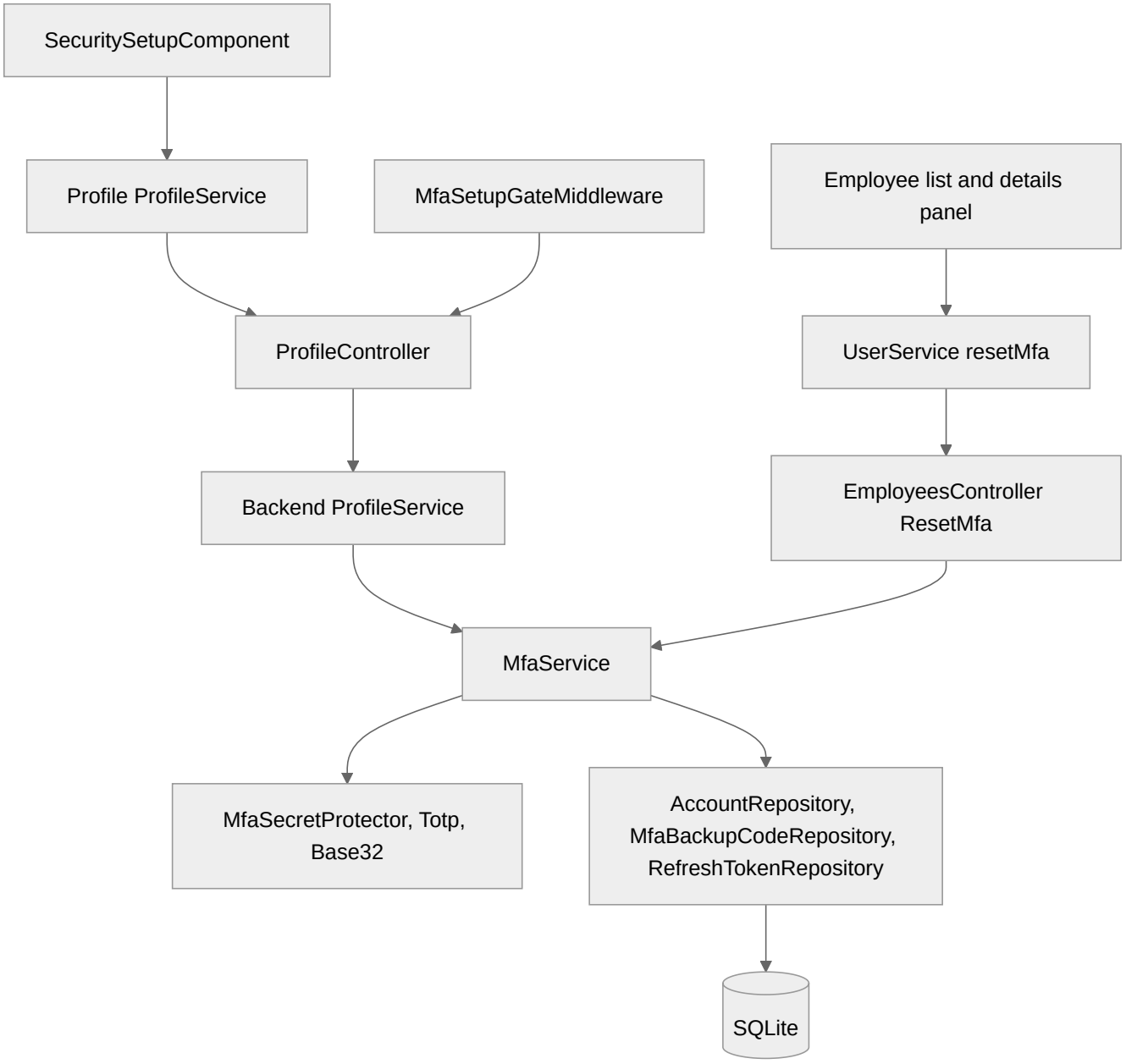
MFA Security Setup – Technical Documentation

Project: TaskFlow – Intelligent Task Allocation & Tracking System
Parent Module: Authentication · **Child Module:** 01.3 MFA Security Setup
Version: 1.0 · **Last Updated:** 31-Oct-2026
Backend: ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core)
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1. Technical Overview

This child implements authenticator-app MFA (TOTP, RFC 6238) in-house (`Integrations/Security/OneTimeCodes/Totp.cs`, `Base32.cs`) and makes enrolment mandatory. Self-service endpoints live on `ProfileController` under `/api/v1/users/me/mfa/*` and call `ProfileService` → `MfaService`; the shared key is encrypted with ASP.NET Core Data Protection (`MfaSecretProtector`) and ten single-use backup codes are stored as SHA-256 hashes. `MfaSetupGateMiddleware` returns `403` with code `mfa_setup_required` for every API call except the setup allow-list while `Mfa:Required` is true and the user has no MFA. The administrator reset is `EmployeesController.ResetMfa` → `MfaService.ResetAsync`. `MfaService.VerifyCoreAsync` is also the code check used by 01.1 Login, 01.2 Account Recovery and Profile. On the client, `SecuritySetupComponent` runs details → authenticator → backup codes, guarded by `securitySetupGuard`; the shell is guarded by `mfaSetupGuard`.

2. Architecture Flow



3. Backend File Structure

All paths are under `TaskFlow Proj/backend/src/TaskFlow.Api/`.

Layer	Path	Files
Controller (MFA self-service, lives in Profile)	`Controllers/Profile/`	`ProfileController.cs` (actions `SetupMfa`, `EnableMfa`, `CheckMfa`, `RegenerateBackupCodes`)
Controller (admin MFA reset, lives in Management)	`Controllers/Management/`	`EmployeesController.cs` (action `ResetMfa`)
Service – MFA	`Services/Auth/Mfa/`	`IMfaService.cs`, `MfaService.cs`

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Layer	Path	Files
Service (wrapper used by Profile MFA endpoints)	`Services/Profile/`	`ProfileService.cs` (`BeginMfaSetupAsync`, `EnableMfaAsync`, `CheckMfaCodeAsync`, `RegenerateBackupCodesAsync`)
Service (used)	`Services/Auth/AttemptLimits/`	`IAttemptLimiter.cs`, `AttemptLimiter.cs` (password re-entry budget, shared, see parent)
Repository	`Repositories/Auth/Mfa/`	`IMfaBackupCodeRepository.cs`, `MfaBackupCodeRepository.cs`
Repository (used)	`Repositories/Auth/Accounts/`, `Repositories/Auth/RefreshTokens/`	`AccountRepository.cs`, `RefreshTokenRepository.cs`
DTOs	`Dtos/Auth/Mfa/`	`MfaRequests.cs` (`MfaSetupRequest`, `MfaSetupResponse`, `MfaEnableRequest`, `MfaPasswordCodeRequest`, `MfaBackupCodesResponse`), `MfaResults.cs` (`MfaVerifyOutcome`, `MfaSetupResult`)
Validators	`Dtos/Auth/Mfa/Validators/`	`MfaRequestValidators.cs` (`MfaSetupRequestValidator`, `MfaEnableRequestValidator`, `MfaPasswordCodeRequestValidator`)
Integrations	`Integrations/Security/MfaSecrets/`	`IMfaSecretProtector.cs`, `MfaSecretProtector.cs` (`Protect`)
Integrations	`Integrations/Security/OneTimeCodes/`	`Totp.cs`, `Base32.cs`
Entities	`Models/Entities/`	`MfaBackupCode.cs`, `User.cs`
Enums	`Dtos/Auth/Mfa/MfaResults.cs`, `Models/Enums/`	`MfaVerifyOutcome` (Success, Invalid, LockedOut); `NotificationType` (System)
Middleware	`Middleware/`	`MfaSetupGateMiddleware.cs`
Authorization (used)	`Authorization/Policies/`	`RequirePermissionAttribute.cs` (`PermissionKeys.EmployeesMfaReset`)
Config	`Config/`	`MfaOptions.cs`

4. Frontend File Structure

All paths are under `TaskFlow Proj/frontend/src/app/`.

Type	Path	Purpose
Routes	`modules/auth/auth.routes.ts`	`security-setup` route with `canActivate: [authGuard, securitySetupGuard]`
Page component	`modules/auth/security-setup/pages/security-setup.component.ts`	Details → authenticator → backup codes (uses `qrcode` package)

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Type	Path	Purpose
Model	`modules/auth/security-setup/pages/security-setup.model.ts`	`SetupStep`, `StepInfo`
API service	`modules/profile/services/profile.service.ts`	`get`, `update`, `startMfaSetup`, `enableMfa`, `checkMfa` (calls `/api/v1/users/me…`)
Models	`shared/auth/auth.models.ts`	`MfaSetup`, `MfaBackupCodes`
Route / Guard	`shared/auth/auth.guard.ts`	`mfaSetupGuard` (shell → `/security-setup` while `mfaSetupRequired`), `securitySetupGuard` (setup page → dashboard once set up)
Interceptor	`shared/auth/auth.interceptor.ts`	On `403` with `code` = `mfa_setup_required` calls `markMfaRequired()` and navigates to `/security-setup`
Admin reset – page	`modules/management/employee-management/pages/employee-list.component.ts` / `.html`	`canResetMfa()` (`Permissions.EmployeesMfaReset`), `resetMfa()` with confirm dialog and toasts
Admin reset – panel	`modules/management/employee-management/components/employee-details-panel.component.ts`	**Two-step** fact and **Reset MFA** button (`canResetMfa` input, `resetMfa` output)
Admin reset – API service	`modules/management/employee-management/services/user.service.ts`	`resetMfa(id)` → `POST /api/v1/users/{id}/mfa/reset`
Permission keys	`shared/auth/permissions.ts`	`EmployeesMfaReset: 'employees.mfa.reset'`

> ⚠️ To be confirmed: `security-setup.component.ts` references `./security-setup.component.css`, but the CSS file is not present in the snapshot.

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5. API & Logic

5.1 API Endpoints

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
6	POST	`/api/v1/users/me/mfa/setup`	Start enrolment after the current password; returns key and `otpauth://` URI once	`[Authorize]` (class-level on `ProfileController`); key `profile.mfa.manage` is Always On, not checked by attribute	`MfaSetupRequest`	`MfaSetupResponse` / `400` / `404` / `429`
7	POST	`/api/v1/users/me/mfa/enable`	Confirm with the first TOTP code; turns MFA on; returns backup codes once	`[Authorize]`; `profile.mfa.manage` (Always On)	`MfaEnableRequest`	`MfaBackupCodesResponse` / `400` / `404`

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#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
8	POST	`/api/v1/users/me/mfa/check`	Check a TOTP code without spending it (Profile "show password")	`[Authorize]`; `profile.mfa.manage` (Always On)	`MfaEnableRequest`	`204` / `400` / `429`
9	POST	`/api/v1/users/me/mfa/backup-codes`	Regenerate backup codes (password + code)	`[Authorize]`; `profile.mfa.manage` (Always On)	`MfaPasswordCodeRequest`	`MfaBackupCodesResponse` / `400` / `404` / `429`
10	POST	`/api/v1/users/{id:int}/mfa/reset`	Administrator reset of another user's MFA; ends their sessions	`[RequirePermission(PermissionKeys.EmployeesMfaReset)]` → policy `perm:employees.mfa.reset`	–	`204` / `404` / `403`

Security Setup step 1 also calls `GET /api/v1/users/me` and `PUT /api/v1/users/me` (`UpdateOwnProfileRequest`), documented in the Profile module.

5.2 Request / Response Samples

```
// POST /api/v1/users/me/mfa/setup
{ "currentPassword": "S3cret!" }
```

```
// 200 OK
{
  "key": "JBSWY3DPEHPK3PXPJBSWY3DPEHPK3PXP",
  "provisioningUri": "otpauth://totp/TaskFlow:jane.doe%40taskflow.com?
secret=JBSWY3DPEHPK3PXPJBSWY3DPEHPK3PXP&issuer=TaskFlow&algorithm=SHA1&digits=6&period=30"
}
```

```
// POST /api/v1/users/me/mfa/enable
{ "code": "123456" }
```

```
// 200 OK – 10 codes, shown once
{ "codes": ["ABCDE-FGHIJ", "KLMNO-PQRST", "..."] }
```

```
// 400 Bad Request (BusinessRuleException via GlobalExceptionHandler)
{ "status": 400, "detail": "The code is incorrect or has expired." }
```

5.3 Service Layer Logic

Method: `ProfileService.BeginMfaSetupAsync(int userId, string currentPassword)` → `MfaService.BeginSetupAsync`

1. `AcquirePasswordAttempt(userId)` – `IAttemptLimiter.TryAcquire("pwd:{id}", null)`; refused → `RateLimitedException("Too many attempts. Try again in a few minutes.")` (`429`).
2. `FindLiveForUpdateAsync(userId)`; null → `404`.

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3. `RequirePassword` – wrong/empty → `BusinessRuleException("Current password is incorrect.")`.
4. MFA already on → `BusinessRuleException("Authenticator MFA is already on. Turn it off first to set it up again.")`.
5. `Totp.NewSecret()` (20 random bytes), `MfaSecretProtected = IMfaSecretProtector.Protect(secret)`, resets `MfaLastUsedStep`, `MfaFailedAttempts`, `MfaLockedUntil`; save.
6. Returns `MfaSetupResult(Base32.Encode(secret), Totp.ProvisioningUri(Mfa:Issuer, user.Email, secret))`. On success the password budget is reset.

Method: `MfaService.EnableAsync(int userId, string code)`

1. Already on → "Authenticator MFA is already on."; no stored key → "Start the setup first to get a key."
2. `VerifyAsync(user, code, allowBackupCode: false)`; `RequireOutcome` throws `LockedMessage` or `InvalidCodeMessage`.
3. Sets `MfaEnabledAt = now`, `ReplaceBackupCodesAsync` (deletes old rows, creates `Mfa:BackupCodeCount` codes: 7 random bytes → Base32 → first 10 chars, shown as `XXXXX-XXXXX`, stored as SHA-256 hex of the 10-char form), save, return codes.

Method: `MfaService.RegenerateBackupCodesAsync(userId, currentPassword, code)` – password attempt budget, password check, requires MFA on ("Authenticator MFA is not on."), `VerifyAsync(allowBackupCode: true)`, replaces codes.

Method: `MfaService.CheckCodeAsync(userId, code)` – MFA off → `Invalid`; otherwise `VerifyCoreAsync(allowBackupCode: false, consumeCode: false, countFailure: true)`.

Method: `MfaService.RequireCodeAsync(userId, code)` – step-up used by Profile password change; no-op if MFA off; empty → "Enter the code from your authenticator app."; otherwise verify with backup codes allowed.

Core: `MfaService.VerifyCoreAsync(user, code, allowBackupCode, consumeCode, countFailure)`

1. `MfaLockedUntil > now` → `LockedOut`.
2. 6 ASCII digits → `Totp.TryValidate(secret, code, now, consumeCode ? MfaLastUsedStep : 0)` (window current step ±1, constant-time compare, step must be greater than last used); on success stores the step.
3. Else, if allowed, `NormalizeBackupCode` (upper-case, strip `-` and spaces, 10 Base32 chars) → `FindUnusedForUpdateAsync(userId, hash)` → sets `UsedAt`.
4. Accepted → reset `MfaFailedAttempts` and lock; `TrySaveAsync` (a `ConcurrencyConflictException` turns success into `Invalid`, so a code cannot be spent twice in parallel).
5. Refused → if `countFailure`, increment `MfaFailedAttempts`; at `Mfa:MaxFailedAttempts` set `MfaLockedUntil = now + Mfa:LockoutMinutes` and reset the counter.

Method: `MfaService.ResetAsync(int userId)` (admin)

1. `FindLiveForUpdateAsync`; null → `false` (`404`).
2. Clears key, `MfaEnabledAt`, counters, lock; removes all backup codes.
3. Rotates `SecurityStamp`, revokes all active refresh tokens, save.
4. Publishes `NotificationType.System`: "An administrator reset your authenticator MFA. Sign in with your password and set it up again from Profile."

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Gate: `MfaSetupGateMiddleware.InvokeAsync` – for authenticated requests carrying the `mfa_setup_required` claim on `/api` paths, anything other than `/api/v1/auth/*`, `/api/v1/users/me/mfa/*` and exactly `/api/v1/users/me` returns `403` ProblemDetails with `detail` "Set up two-step verification to continue." and `code` = `mfa_setup_required`. The claim is added by `SessionValidation` (01.4).

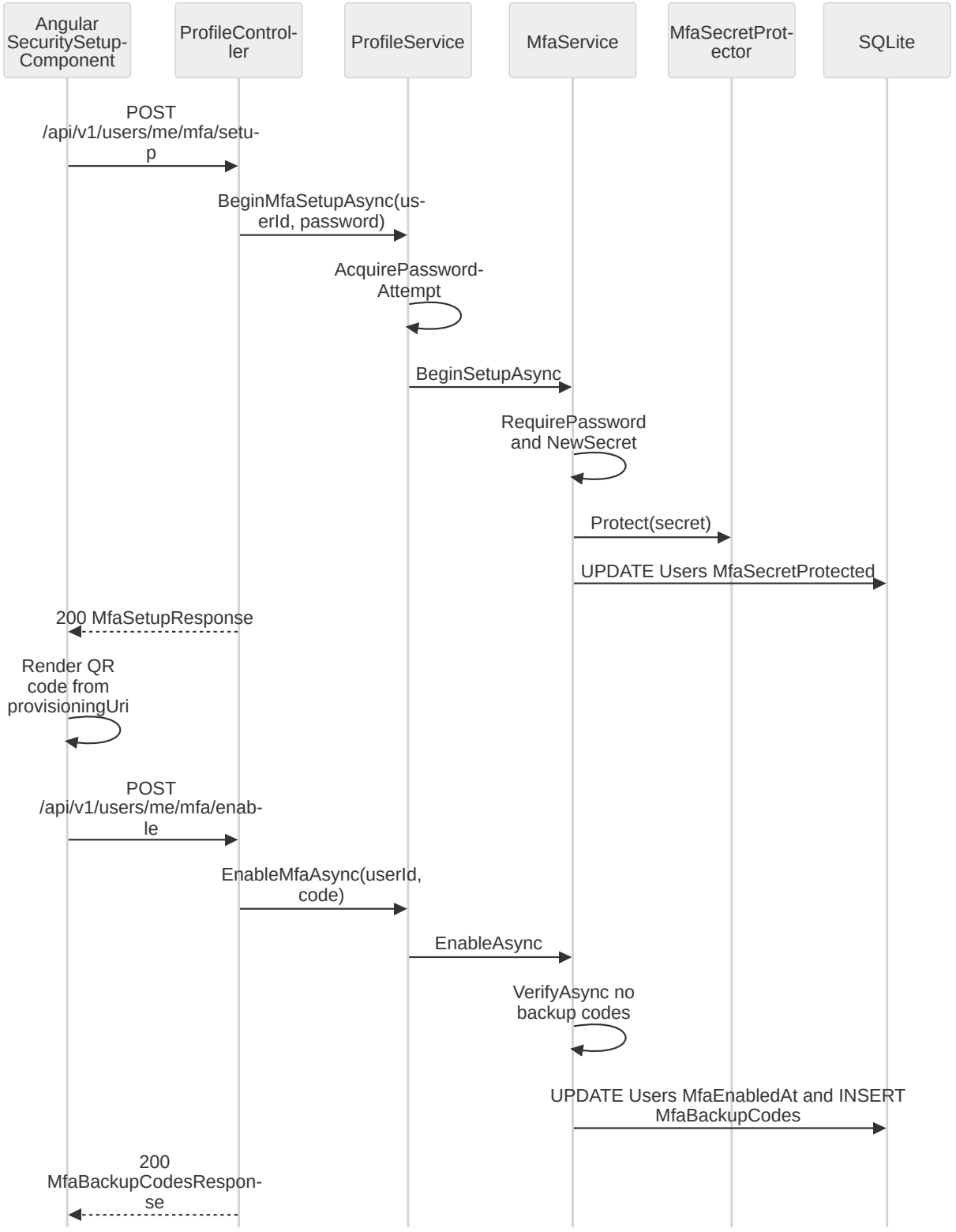
Frontend admin reset: `EmployeeListComponent.resetMfa(employee)` opens the confirm dialog ("Reset two-step verification?", confirm label "Reset MFA", variant danger), calls `UserService.resetMfa(id)`, then shows the toast "MFA reset for (name)" and sets `mfaEnabled: false`, `backupCodesRemaining: 0` in the panel; on error the toast is `apiErrorMessage(err, 'Could not reset MFA.')`.

5.4 Repository Layer

Method	Description	Query Notes
`AccountRepository.FindLiveForUpdateAsync(userId)`	User by id, not deleted	Tracked
`MfaBackupCodeRepository.FindUnusedForUpdateAsync(userId, codeHash)`	Matching unused backup code	Tracked; `UsedAt == null`; uses index `(UserId, CodeHash)`
`MfaBackupCodeRepository.GetForUserForUpdateAsync(userId)`	All of the user's codes	Tracked; used to delete before re-creating
`RefreshTokenRepository.GetActiveForUsersForUpdateAsync(userIds)`	Active tokens for revocation (admin reset)	Tracked

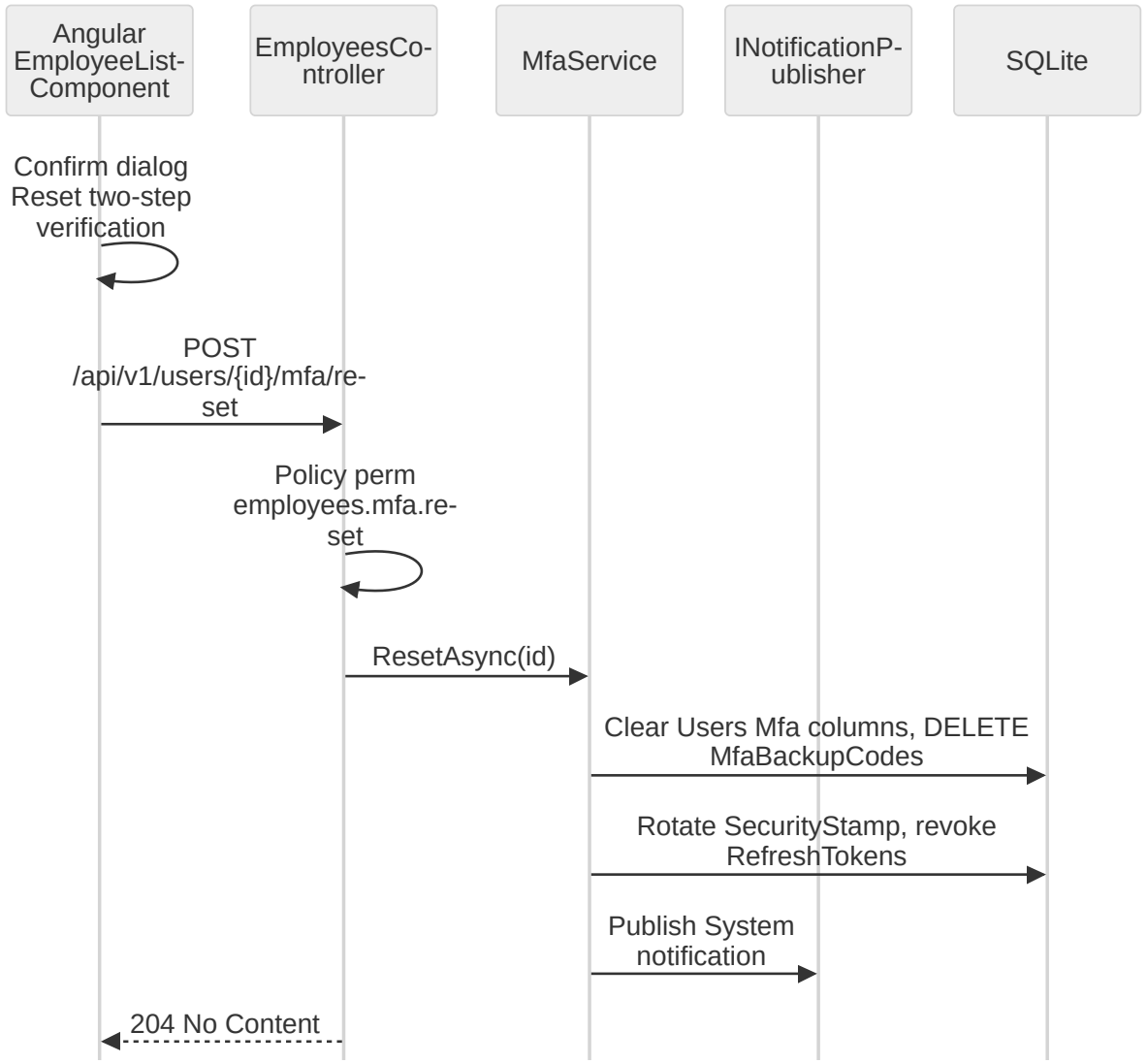
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5.5 Sequence Diagram



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Administrator reset



6. Database Design

This child owns `MfaBackupCodes` and the `Users.Mfa*` columns. Full `Users` column list and relationships: parent document section 6.

Table	Column	Type	Constraints	Description
`Users`	`MfaSecretProtected`	TEXT(512)	Nullable	Data-Protection-encrypted TOTP key
`Users`	`MfaEnabledAt`	TEXT (DateTime)	Nullable	When MFA was turned on
`Users`	`MfaLastUsedStep`	INTEGER (long)	Not Null, concurrency token	Last accepted TOTP step (replay protection)
`Users`	`MfaFailedAttempts`	INTEGER	Not Null, default 0, concurrency token	Consecutive wrong codes

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Table	Column	Type	Constraints	Description
`Users`	`MfaLockedUntil`	TEXT (DateTime)	Nullable	MFA lockout end
`Users`	`PasswordHash`	TEXT	Not Null	Checked before setup and backup-code regeneration
`Users`	`SecurityStamp`	TEXT(64)	Not Null	Rotated on admin MFA reset
`Users`	`MobileNumber`	TEXT(20)	Nullable	E.164 number (collected on Security Setup step 1 through Profile)
`MfaBackupCodes`	`Id`	INTEGER	PK	–
`MfaBackupCodes`	`UserId`	INTEGER	FK → `Users.Id`	Owner
`MfaBackupCodes`	`CodeHash`	TEXT(64)	Not Null; index `IX_MfaBackupCodes_UserId_CodeHash`	SHA-256 hex of the normalised code
`MfaBackupCodes`	`CreatedAt`	TEXT (DateTime)	Not Null	–
`MfaBackupCodes`	`UsedAt`	TEXT (DateTime)	Nullable, concurrency token	When spent
`RefreshTokens`	`RevokedAt`	TEXT (DateTime)	Nullable	Set for every active token of the user on admin reset (columns in 01.4)

Relationships

- `MfaBackupCodes.UserId → Users.Id` (FK, Cascade)

7. Security & Authorization

- **MFA:** TOTP SHA1 / 6 digits / 30 s, ±1 step, single use per step; lockout after 5 wrong codes for 5 minutes; key encrypted with Data Protection (purpose `TaskFlow.MfaSecrets`, key ring under `AiProviders:KeyRingPath`, DPAPI-protected on Windows).
- **Mandatory enrolment:** `MfaSetupGateMiddleware` runs after `UseAuthentication` and before `UseAuthorization`; allow-list `/api/v1/auth/*`, `/api/v1/users/me/mfa/*`, exactly `/api/v1/users/me`.
- **Password re-entry:** setup and backup-code regeneration need the current password, throttled by `AttemptLimiter` key `pwd:{id}` (5 per 5 minutes).
- **Data scope:** Own – every `/users/me` MFA endpoint takes the user id from the token (`ClaimTypes.NameIdentifier`), never from the request. Organization – `employees.mfa.reset` applies to any user id.
- **Frontend:** `mfaSetupGuard` on the shell, `securitySetupGuard` on `/security-setup`; the **Reset MFA** button in Employee Management is shown only with `employees.mfa.reset` and when the employee has MFA on.

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Permission Key	Endpoint(s)	UI Element Controlled
`profile.view` (Always On, not attribute-checked)	`GET /api/v1/users/me`	Security Setup step 1 facts
`profile.edit` (Always On, not attribute-checked)	`PUT /api/v1/users/me`	Security Setup step 1 form
`profile.mfa.manage` (Always On, not attribute-checked)	`POST /api/v1/users/me/mfa/setup`, `/enable`, `/check`, `/backup-codes`	Security Setup steps 2–3
`employees.mfa.reset` (Administrator Granted; Manager/Employee Denied; Custom Configurable)	`POST /api/v1/users/{id}/mfa/reset`	**Reset MFA** button in the employee details panel

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
`CurrentPassword` required	`MfaSetupRequestValidator`, `MfaPasswordCodeRequestValidator`	400	FluentValidation defaults
`Code` required, max 32	`MfaEnableRequestValidator`, `MfaPasswordCodeRequestValidator`	400	FluentValidation defaults
Wrong current password	`MfaService.RequirePassword`	400	Current password is incorrect.
Too many password re-entries	`ProfileService.AcquirePasswordAttempt`	429	Too many attempts. Try again in a few minutes.
MFA already on (setup)	`MfaService.BeginSetupAsync`	400	Authenticator MFA is already on. Turn it off first to set it up again.
MFA already on (enable)	`MfaService.EnableAsync`	400	Authenticator MFA is already on.
No pending key	`MfaService.EnableAsync`	400	Start the setup first to get a key.
MFA not on (backup codes)	`MfaService.RegenerateBackupCodesAsync`	400	Authenticator MFA is not on.
Wrong TOTP / backup code	`MfaService.RequireOutcome`, `ProfileController.CheckMfa`	400	The code is incorrect or has expired.
MFA locked	`ProfileController.CheckMfa` / `MfaService`	429 / 400	Too many wrong codes. Try again in a few minutes. (shared rule, see parent)
Step-up code missing	`MfaService.RequireCodeAsync`	400	Enter the code from your authenticator app.
MFA setup pending	`MfaSetupGateMiddleware`	403	Set up two-step verification to continue. (`code`: `mfa_setup_required`)
No permission (`employees.mfa.reset`)	Policy `perm:employees.mfa.reset`	403	Forbidden

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Rule	Layer	HTTP Code	Error Message
User not found (MFA endpoints / admin reset)	Controller	404	(empty body)

9. Integrations / Side Effects

- **Notifications:** `INotificationPublisher.PublishAsync(..., NotificationType.System, ...)` – administrator MFA reset → "An administrator reset your authenticator MFA. Sign in with your password and set it up again from Profile."
- **Email:** none.
- **Activity Log:** none; a denied `employees.mfa.reset` call is logged by `DeniedActionAuditHandler`.
- **Session effects:** admin MFA reset rotates `SecurityStamp` and revokes all refresh tokens (all sessions of the employee end).
- **Profile:** `RequireCodeAsync` is the step-up check for the Profile password change; `CheckCodeAsync` backs the Profile "show password" check.

10. Configuration

Key (`appsettings.json`)	Purpose	Default
`Mfa:Issuer`	Name shown in the authenticator app	`TaskFlow`
`Mfa:BackupCodeCount`	Backup codes per set (code default)	`10`

`Mfa:Required`, `Mfa:MaxFailedAttempts`, `Mfa:LockoutMinutes` and `AiProviders:KeyRingPath` are shared and listed in the parent document.

11. Testing

Paths are relative to `backend/tests`.

Test Type	File / Location	Covers
Unit	`TaskFlow.UnitTests/Auth/Mfa/MfaServiceTests.cs`	Enrolment, verification, lockout, backup codes, reset
Unit	`TaskFlow.UnitTests/Integrations/Security/OneTimeCodes/TotpTests.cs`	TOTP generation and validation
Integration	`TaskFlow.IntegrationTests/Auth/Mfa/MfaEnrolmentEndpointsTests.cs`	`/users/me/mfa/*` enrolment
Integration	`TaskFlow.IntegrationTests/Auth/Mfa/MfaSetupGateEndpointsTests.cs`	`MfaSetupGateMiddleware`
Integration	`TaskFlow.IntegrationTests/Auth/Mfa/MfaStepUpEndpointsTests.cs`	Step-up code (`RequireCodeAsync`)
Integration	`TaskFlow.IntegrationTests/Auth/Mfa/MfaAdminResetEndpointsTests.cs`	`POST /users/{id}/mfa/reset`

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Shared: `MfaTestSupport.cs`, `AcceptingMfaService.cs` and `AttemptLimiterTests.cs` are listed in the parent document. No frontend spec files were found under `modules/auth/security-setup`.

12. Known Limitations / Gaps found

- There is no user-facing endpoint to turn MFA off, although `MfaSetupRequest` / `MfaPasswordCodeRequest` comments and the "Turn it off first" message mention it; only the administrator reset removes MFA.
- **Regenerate backup codes:** `POST /api/v1/users/me/mfa/backup-codes` exists but no UI calls it (`ProfileService` frontend has no method for it; the permission audit reports the same). `profile.mfa.manage` describes "regenerate backup codes".
- **Admin-reset texts:** the admin-reset notification says "set it up again from Profile", but setup happens on `/security-setup`; the permission audit notes there is no MFA card on Profile. The confirm dialog in `employee-list.component.ts` also says the employee "can sign in with just their password", while with `Mfa:Required` = true the gate sends them to Security Setup.
- **Post-setup redirect:** `SecuritySetupComponent.finish()` calls `dashboardRouteFor(role)` without permissions, so custom roles with organization/team dashboard permissions land on `/me` (login uses the permissions correctly).
- **Always-on keys not enforced as policies:** `profile.view`, `profile.edit`, `profile.mfa.manage` are Always On and not stored, so `EffectivePermissions.Expand` never adds them as claims; the endpoints rely on `[Authorize]` only. This matches the matrix (`Always On`) but means `RequirePermission(profile.*)` must not be used.
- **Missing style file:** `security-setup.component.css` is referenced but absent from the snapshot.

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